

Unmanned Autonomous Surface Vehicle with LiDAR and Vision Based Collision Avoidance for Offshore Renewable Energy Exploration

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Abstract—Offshore renewable energy surveys in Indonesia are still carried out with manned vessels and heavy equipment. This approach is costly and slow, and it puts the crew at risk when sea conditions change quickly. This paper describes a catamaran Unmanned Autonomous Surface Vehicle (UASV) built for such surveys. The vehicle follows a waypoint route by itself and avoids obstacles using data from a LiDAR and a camera. It carries an RPLiDAR A2M12 for short-range measurement, a camera running YOLOv8n on a Jetson Nano with TensorRT for obstacle recognition at longer range, and a HERE3 RTK GPS for positioning. The subsystems are connected through ROS nodes and a Pixhawk 2.4.8 flight controller. Field trials were carried out at Gili Ketapang, East Java. RTK navigation produced a mean position error of 0.2129 m and a waypoint tracking error below 0.5 m. After TensorRT conversion the detector ran at 29.8 FPS with a precision of 0.989 and an F1-score of 0.9945, which is 1.68 times faster than the unconverted YOLOv8n at a comparable accuracy. The avoidance module recognised a fishing boat and a buoy about 2 m ahead, kept a clearance of 0.5–0.8 m, and returned to the planned track with a maximum lateral deviation of 1.2 m over a 22 m waypoint course in shallow water near the shore. Endurance at full load was 35–40 minutes. During the same mission the vehicle also recorded water depth, wind speed and direction, and coral positions. The results are a functional field validation at small scale and indicate that a vehicle of this size is a workable basis for preliminary surveys of offshore energy sites.

Index Terms—unmanned surface vehicle, collision avoidance, sensor fusion, YOLOv8, TensorRT, RTK GPS, LiDAR, offshore renewable energy

I. INTRODUCTION

Energy consumption worldwide continues to grow. In 2018 primary energy consumption rose by 2.9% compared with the previous year, the fastest rate since 2010, while carbon emissions from energy use grew by 2.0%, an increase of about 0.6 gigatonnes [1]. Indonesia is in a good position to supply part of this demand from the sea. Around two thirds of its

territory is water, and its coastline is the second longest in the world. Tidal energy, wave energy, ocean currents, offshore wind, and ocean thermal energy conversion (OTEC) can all be developed within its waters.

The difficulty lies in the survey stage rather than in the resource itself. Conventional offshore exploration uses manned vessels and heavy equipment, and it needs a large crew to stay at sea for long periods. Operating costs are high, the crew is exposed to risk when sea conditions are severe, and fuel consumption is large. At the same time, data on bathymetry, seabed condition, wind, and currents must be available before foundations for wind turbines or floating solar platforms can be designed, and before installations can be placed where the energy yield is high and the ecological impact is small.

Sea conditions off the Indonesian coast are hard to predict. Waves can be high, currents can be strong, the weather changes suddenly, and small-craft traffic in coastal areas is dense. Survey equipment for these waters therefore has to operate on its own and has to avoid obstacles without an operator. This paper presents an Unmanned Autonomous Surface Vehicle (UASV) with a collision avoidance system intended for offshore energy exploration. The main contributions are as follows.

- 1) A modular catamaran UASV with an aluminium frame and balsa hulls, driven by four thrusters in an omni-directional layout and analysed hydrostatically before fabrication.
- 2) A collision avoidance scheme in which the LiDAR handles obstacles close to the hull, up to about 12 m and without a blind spot, while a YOLOv8n detector converted with TensorRT handles obstacles further away and supplies the object class.
- 3) Field measurements in the coastal waters of Gili Ketapang, Probolinggo, including a comparison of LiDAR

ranging accuracy on land and on water, and a benchmark of four YOLO nano variants on the Jetson Nano.

- 4) A web-based control system built with Flask that brings telemetry, video streaming, waypoint dispatch, and data logging into a single operator interface.

II. RELATED WORK

Several low-cost autonomous surface vehicles for coastal monitoring have been reported. Carlson *et al.* built a portable ASV with obstacle avoidance for coastal ocean monitoring [2]. Campos *et al.* developed *Nautilus*, which uses a multilayer software architecture for offshore inspection [3]. Both show that small platforms are suitable for survey work, although neither collects the combination of site parameters needed for renewable energy assessment.

Work on collision avoidance for USVs has taken several directions. Yuan *et al.* adapted the Dynamic Window Approach to COLREGs and included the effect of wind and waves [4]. Sun *et al.* used deep reinforcement learning together with navigation characteristics [5], and Yun *et al.* combined a kinodynamic RRT variant with model predictive control based on a radial basis function network so that actuator faults could be tolerated [6]. Barrera *et al.* compared several intelligent path planning models for environmentally propelled USVs [7], and Jeong and Quattrini Li released MARCOL, a testbed for maritime collision avoidance decisions [8]. Most of this work is concerned with the planner. The present study is concerned instead with running a fused perception front end in real time on low-power embedded hardware, and with testing it in the field rather than in simulation.

In terms of perception, Argeze examined sensor selection and combined LiDAR with YOLOv4 for an ASV in the Venice Lagoon [9]; that arrangement is the closest to the one used here. Compact detector architectures such as YOLO Nano [10], together with reported work on deep networks for visual localisation on constrained hardware [11], support the use of YOLOv8n with TensorRT quantisation on the Jetson Nano.

III. SYSTEM DESIGN

A. Mechanical Design

A catamaran configuration was chosen for stability and for efficient operation in changing sea conditions (Fig. 1). The frame is made of aluminium because it combines strength, low weight, and resistance to corrosion. The two hulls are made of balsa coated with epoxy resin, which keeps the displacement low while protecting the structure against seawater and light impact. The structure is modular. The main frame, the sensor mast, and the thruster mounts are built as separate parts and joined with stainless steel bolts, so the vehicle can be taken apart for transport, maintained one module at a time, and reconfigured for different missions. Heavy components such as the batteries and the thrusters are placed low so that the centre of gravity stays low.

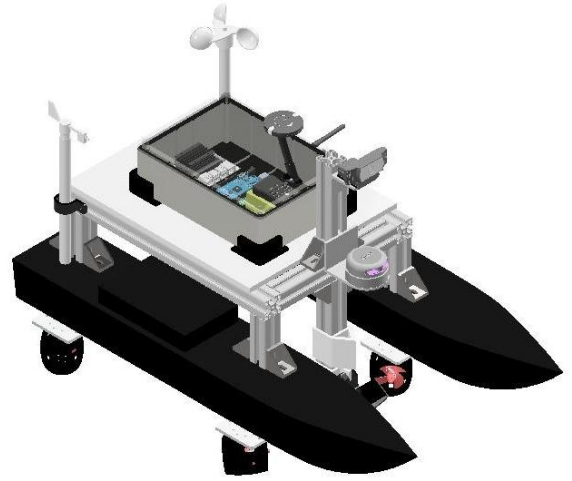


Fig. 1. Three-dimensional design of the catamaran UASV.

Table I lists the principal dimensions, which were obtained from a Maxsurf analysis carried out before fabrication.

TABLE I
PRINCIPAL PARTICULARS OF THE UASV

Parameter	Value
Length overall (LOA)	90 cm
Length on waterline (LWL)	82.73 cm
Length between perpendiculars (LPP)	75 cm
Draft (T)	8 cm
Beam (B)	44.4 cm
Height (H)	48.5 cm
Displaced volume	7 231.19 cm ³
Displacement	7.411 kg
Wetted area	3 121.28 cm ²
Prismatic coefficient (C_p)	0.829

Propulsion is provided by four brushless thrusters mounted at 45° to the longitudinal axis in an Omni-X arrangement, which gives three degrees of freedom: surge along x , sway along y , and yaw about z . Brushless motors were preferred over brushed motors because they are more efficient, tolerate salt water better, and can be controlled more precisely by an electronic speed controller. The thrusters were positioned so that the load is distributed evenly about the centre of mass. A thruster placed too close to the centre of mass reduces the yaw moment and makes lateral movement harder to control.

The sensors were placed so that they interfere with each other as little as possible. The camera faces forward with a wide field of view. The LiDAR sits on a corrosion-resistant mast above the main frame so that its view is not blocked. The RTK GPS is mounted on the rear frame, away from the electromagnetic noise of the thrusters. The IMU is at the centre of the frame, the depth sensor is under the hull, and the anemometer is at the stern.

B. Hardware Architecture

Three processors form the core of the electronics (Fig. 2). An ATmega2560 reads the low-level sensors and handles the serial control links over UART, I²C, and SPI. A Jetson Nano

runs the vision pipeline, that is YOLOv8n under TensorRT, and also acts as the link between the microcontroller and the telemetry system. A Pixhawk 2.4.8 carries out thruster mixing and attitude stabilisation using its MPU6000 IMU, which keeps the vessel on heading in a seaway.

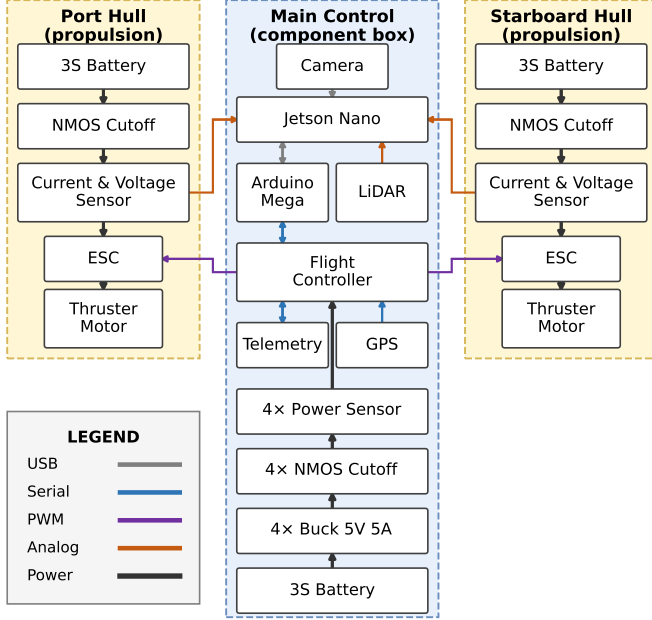


Fig. 2. Communication architecture of the UASV. Line colour denotes the link type; the power distribution chain is shown in black.

The perception hardware consists of a Logitech HD Pro C920 camera, an RPLiDAR A2M12 scanning rangefinder, a HERE3 RTK GPS paired with a shore base station, and the MPU6000 IMU. Power comes from a LiPo pack and is distributed through a power distribution board to the thrusters and to the computing and sensing payload.

C. Software Architecture

The software runs on ROS, so that perception, navigation, and actuation can be written as separate nodes (Fig. 3):

- the *remote node* reads the RC receiver and converts the signal to PWM for the flight controller;
- the *vision node* performs object detection and labelling and estimates the position of an object relative to the vehicle;
- the *LiDAR node* converts raw scans into range and bearing reports;
- the *MAVROS node* handles MAVLink communication with the flight controller;
- the *navigation node* combines data from the remote, vision, LiDAR, and MAVROS nodes into navigation decisions and thruster commands.

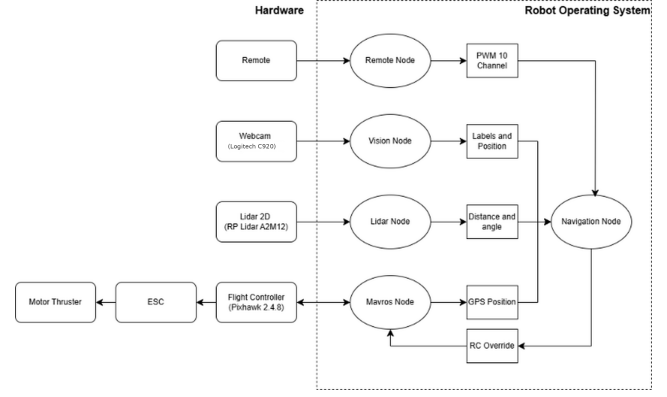


Fig. 3. ROS node graph of the UASV software stack.

Communication with the ground control station is two-way and uses MAVLink, so that status telemetry and commands are exchanged while the mission is running.

D. Collision Avoidance Strategy

The two sensors are used for different parts of the problem rather than as backups for each other.

1) *LiDAR for the near field*: The RPLiDAR A2M12 scans a sector of 90° to 270° and measures range up to about 12 m. It does not depend on ambient light, and it covers the area immediately around the hull where the camera cannot see. Its drawbacks are the limited range and the fact that it gives no information about what the obstacle is.

2) *Vision for the far field*: The camera with YOLOv8n detects and classifies objects beyond the LiDAR range and reports shape, size, and class (boat, head, buoy). The planner is therefore warned earlier and can tell a moored buoy apart from an approaching fishing boat. The camera is, however, affected by low light and bad weather.

3) *Fusion and avoidance*: Detections from both sensors are combined in the navigation node. If an obstacle falls on the planned track, the node produces a new trajectory by changing heading, speed, or both, keeps a safety margin while passing the obstacle, and then returns to the waypoint line. Safety is given a higher priority than path length, and a detection reported by only one sensor is checked against the other before it is acted on.

E. Dataset and Detector Training

The images were collected by operating the UASV manually in the area where it was to be used, at different times of day and in clear, cloudy, and rainy weather. Annotation was done in Roboflow for three classes: *boat*, *head* (bow), and *buoy*. Augmentation was limited to horizontal flipping, random cropping, and small rotations, because these reproduce what actually happens when the vehicle rolls in waves and the camera framing is unsteady. Brightness adjustment and noise injection were not used, so that the lighting recorded in the field was preserved. The dataset was divided into 70% for training, 20% for validation, and 10% for testing.

Training used the Ultralytics YOLOv8 implementation with a learning rate of 0.01, a batch size of 16, and 100 epochs.

Validation loss and mAP were checked at every epoch. The trained model was exported to ONNX and then compiled into a TensorRT engine at reduced precision on the Jetson Nano.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

Testing started with bench and pool trials and continued with field missions in the coastal waters of Gili Ketapang, Probolinggo, East Java. Figure 4 shows the Jaladhi UASV during field testing on the water.



Fig. 4. Jaladhi UASV during field testing in coastal waters.

A. Mechanical and Manual-Mode Trials

Balance and leak tests were carried out before the electronics were installed. No leakage was found in the first pool trials, but a longer trial at Danau Graha ITS revealed a crack at a drainage joint in one hull. The area was marked and sealed temporarily, then repaired with filler and fresh resin. The procedure was repeated until the hull remained dry. Manual trials with the RC transmitter were then carried out. The steering and propulsion responded correctly to forward, reverse, and turning commands, and the heading remained stable.

B. Battery Endurance

Endurance was measured during a 45-minute mission at full load with all onboard systems active. Pack voltage was logged at intervals by an Arduino-based monitor (Fig. 5). The voltage was steady at the beginning of the run and then fell gradually as the thruster load increased. The measurements give the following results:

- the vehicle can be operated for 35–40 minutes before the pack reaches the recommended minimum voltage;
- a full recharge takes about 2 hours;
- the largest loads are the thrusters and the Jetson Nano while YOLOv8 inference is running;
- no sudden voltage drop was recorded, which indicates that the pack is in good condition and the load is balanced.

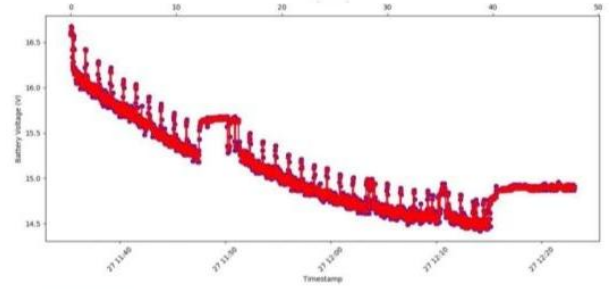


Fig. 5. Battery voltage over a full-load mission.

C. RTK GPS Accuracy and Waypoint Tracking

The HERE3 RTK GPS was tested against a shore base station that supplied corrections for multipath and ionospheric delay. The site was chosen so that no trees, buildings, or other tall structures blocked the sky below an elevation of 30° at the vehicle or at the base station, because signal quality below that angle has a strong effect on the reliability of the RTK fix.

Waypoints were taken from a planned route, and the logged positions were compared with the planned coordinates. The mean position error was 0.2129 m, and the vehicle followed the waypoint line with a mean error below 0.5 m, which satisfies the requirement for autonomous navigation in this environment. The accuracy remained consistent despite differences in terrain and weather during the campaign.

D. LiDAR Ranging: Land versus Water

The ranging accuracy of the LiDAR was measured on land and on water at distances from 0.5 m to 12 m against a reference value (Table II, Fig. 6).

TABLE II
RPLiDAR A2M12 RANGING ERROR ON LAND AND ON WATER

True (m)	LiDAR land (m)	LiDAR water (m)	Error land (m)	Error water (m)
0.5	0.502	0.515	0.002	0.015
1.0	1.009	1.030	0.009	0.030
2.0	2.020	2.050	0.020	0.050
3.0	3.030	3.100	0.030	0.100
4.0	4.050	4.200	0.050	0.200
5.0	5.080	5.350	0.080	0.350
6.0	6.100	6.400	0.100	0.400
8.0	8.150	8.500	0.150	0.500
10.0	10.250	10.600	0.250	0.600
12.0	12.300	12.800	0.300	0.800

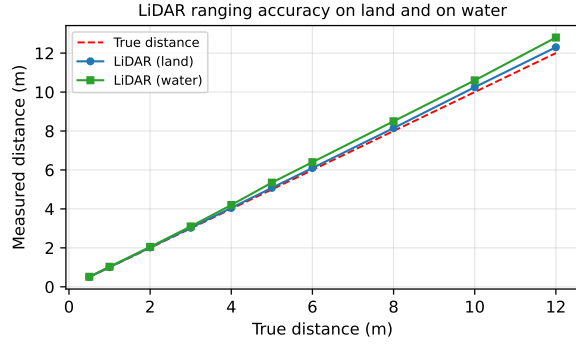


Fig. 6. LiDAR ranging accuracy on land compared with on water.

Up to about 3 m the readings on the two media are almost the same. Beyond that distance the error on water grows faster, and at 12 m it reaches 0.8 m, or 6.7%, compared with 0.3 m, or 2.5%, on land. This is caused by the reflectivity of a moving water surface, which scatters the returns and biases the time-of-flight measurement. The LiDAR should therefore be treated as an accurate ranging source only at short range, which is the role given to it in the fusion scheme, while longer ranges are left to the camera.

E. Detector Benchmark on the Jetson Nano

Four nano-scale YOLO variants were tested on the same dataset and the same hardware (Table III).

TABLE III
COMPARISON OF YOLO NANO VARIANTS ON THE JETSON NANO

Model	Prec.	Rec.	mAP @.5	mAP @.5:.95	F1	FPS
YOLOv8n	0.995	0.999	0.995	0.871	0.997	17.74
YOLOv10n	0.960	0.975	0.987	0.846	0.968	17.83
YOLOv11n	0.986	0.999	0.995	0.838	0.993	15.71
YOLOv8n TRT	0.989	0.999	—	—	0.995	29.8

YOLOv8n compiled with TensorRT gives the best result for this application. At 29.8 FPS it is 1.68 times faster than the unconverted YOLOv8n and 1.90 times faster than YOLOv11n, while precision falls only from 0.9949 to 0.9890 and the F1-score from 0.9974 to 0.9945. mAP is not reported for the TensorRT engine because the deployment toolchain could not evaluate it after conversion. The precision, recall, and F1-score measured on the engine indicate that quantisation did not cost significant accuracy. A rate of about 30 FPS is fast enough for the vehicle to react to obstacles at its survey speed, whereas the variants running at about 17 FPS are not.

F. Web Control System

The ground interface is a Flask application served over the local network of the vehicle or over its own hotspot. It shows geographical data (latitude, longitude, altitude), wind speed and direction from the anemometer, a live video stream from the forward camera, and plots of the sensor data. From the control panel the operator can switch between RC and

autonomous mode, send waypoints, turn the visualisation on and off, and download a CSV log containing position, wind data, and timestamps.

All of these functions were used during the field test at Gili Ketapang. The change of mode was smooth, the waypoints that were sent were followed accurately, the video stream was stable, and the GPS and wind data appeared without delay and agreed with the observed conditions. The Flask server ran for the whole mission without interruption while handling communication, display, and sensor processing at the same time.

G. Environmental Survey

An autonomous survey was carried out in the coastal water west of Gili Ketapang, collecting three parameters at once: depth from an underwater ultrasonic sensor, wind from the anemometer, and the presence of coral from an underwater camera with a YOLO-based detector. The vehicle logged 150 points in total. Nine of these lie on the approach transect running south along the western edge of the area; the remaining 141 form the survey block, which covers about 67 m east to west by 43 m north to south and is the area shown in Fig. 7. Depth was logged at every point and wind at every tenth point.

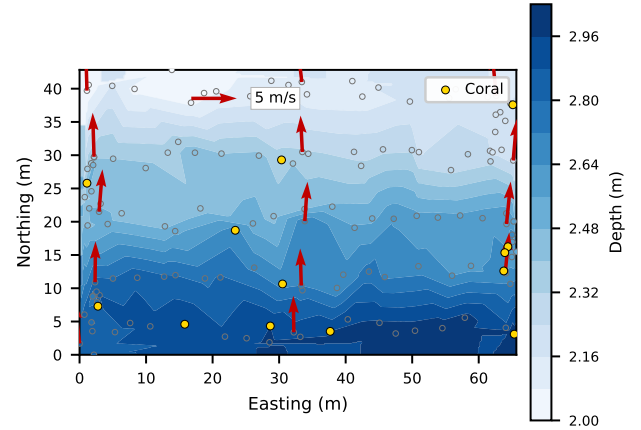


Fig. 7. Survey output: bathymetry, wind vectors, and detected coral locations at Gili Ketapang. Axes are metres east and north of the south-west corner of the survey block, at 7.680921° S, 113.243508° E. Grey dots are logged sample points; the depth field is interpolated between them.

Depth over the block ranges from 2.01 m to 3.04 m with a mean of 2.53 m, shallow enough for early-stage exploration and for shallow-water installations. Wind speed lies between 4.0 m/s and 4.9 m/s with a mean of 4.4 m/s, and the direction is consistently southerly, between 173° and 186°. This is a moderate and steady regime, relevant both to the placement of floating platforms and to the route planning of the vehicle itself. Coral was detected at 13 points within the block, mostly along its margins, which matters for navigation safety as well as for environmental impact assessment.

H. Collision Avoidance Trial at the Survey Site

The avoidance system was tested in the same coastal area, about 560 m south-east of the survey block and roughly 1 km

west of Gili Ketapang island. Three waypoints were laid out along a straight line, and two real obstacles, a fishing boat and a buoy, were placed so that they blocked the track. An evaluation point was used to record the position of the vehicle after the avoidance manoeuvre (Table IV). The course is short: its two legs measure 15.7 m and 6.1 m, so the whole run covers about 22 m. The trial is therefore a functional test of the avoidance pipeline at close range rather than a transit over an extended route.

TABLE IV
TRAJECTORY DATA FOR THE COLLISION AVOIDANCE TRIAL

No	Latitude	Longitude	Description
1	-7.6832962	113.2482636	Waypoint 1
2	-7.6832656	113.2481244	Waypoint 2
3	-7.6832735	113.2480698	Waypoint 3
4	-7.6832650	113.2481183	Obstacle: boat
5	-7.6832777	113.2482329	Obstacle: buoy
6	-7.6832574	113.2481579	Evaluation point

Both obstacles were detected by the camera and the LiDAR at a range of about 2 m before the hazard zone. The vehicle then carried out an avoidance manoeuvre on its own and passed each object with a clearance of 0.5–0.8 m. The largest lateral deviation from the line between waypoints was about 1.2 m, which is acceptable given the scale of the course. After passing the obstacle the vehicle continued to the next waypoint without any manual intervention (Fig. 8).

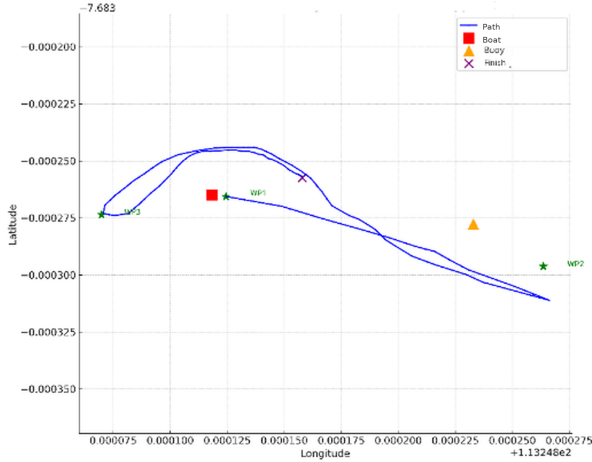


Fig. 8. Recorded trajectory during the collision avoidance trial. The blue line is the executed path; markers indicate the boat, the buoy, and the waypoints.

The trial shows that the combined perception and avoidance pipeline runs in real time on the water, and that the vehicle can decide by itself to deviate and then return to the planned track. This is the behaviour required of a survey platform that has to work in coastal waters used by other vessels, although the short course and the single run mean the result should be read as a functional demonstration rather than as a performance figure.

V. DISCUSSION

The measurements support the way the sensing task was divided between the two sensors. The LiDAR test shows that range accuracy on water degrades beyond about 3 m, and this is the region in which the camera takes over. The TensorRT conversion is what makes the design usable on a Jetson Nano. Without it the detector runs at 17 FPS, which does not leave enough reaction distance at the survey speed, given the detection range of about 2 m observed in the field. The RTK accuracy of 0.2129 m also matters, because it is what allows a deviation of 1.2 m to be read as a controlled manoeuvre rather than as navigation noise.

Several limitations remain. The avoidance trial covered a single run over a 22 m course in shallow water near the shore, so it demonstrates that the pipeline functions but does not establish a success rate, and it does not cover the sea states or the transit distances of an offshore mission. The endurance of 35–40 minutes limits the mission radius. The detection range of about 2 m observed in the field is conservative compared with the nominal capability of the sensors, and it restricts the safe transit speed. All obstacles used in the avoidance trial were stationary, so encounters with moving traffic under COLREGs were not evaluated.

VI. CONCLUSION

An unmanned autonomous surface vehicle with LiDAR and vision based collision avoidance was designed, built, and tested in the field for offshore renewable energy exploration. The platform passed the leak and balance tests and runs for 35–40 minutes at full load without abnormal voltage behaviour. RTK navigation gave a mean position error of 0.2129 m with waypoint tracking below 0.5 m. The YOLOv8n detector converted with TensorRT reached 29.8 FPS at a precision of 0.989 and an F1-score of 0.9945, which is 1.68 times faster than the baseline with only a small loss of accuracy. In the coastal waters of Gili Ketapang the vehicle detected and avoided a fishing boat and a buoy on its own over a 22 m waypoint course, keeping a clearance of 0.5–0.8 m and a maximum track deviation of 1.2 m, and at the same time mapped depth, wind, and coral distribution over the survey grid.

Further work will add cloud storage for navigation, sensor, and video data so that a mission can be analysed remotely after the vehicle is recovered. Testing will also be extended to more difficult conditions, such as high waves, strong currents, and areas with heavy vessel traffic, and to moving obstacles under COLREGs.

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